Session 1: September 2, 2025 (Tuesday, 10:00 am – 11:30 am)

- Course introduction and assignments explanation
- How to write Specific Aims page

Session 2: September 23, 2025 (Tuesday, 9:00 am – 11:00 am)

- How to write Research Strategy and Candidate Information

Session 3: November 18, 2025 (Tuesday, 9:00 am – 11:30 am)

- Presentation and peer discussion
- Each student has 10 minutes for presentation

Session 4: December 9, 2025 (Tuesday, 10:00 am – 11:30 am)

- Final class presentation
- Each student has 10 minutes for presentation

INSTRUCTORS

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COURSE DESCRIPTION

This course will lead students through the process of writing a Master's Essay in the form of an NIH-style grant application (required for the Master of Science in Patient Oriented Research program). The essay is undertaken during the fall semester of the second year of study. At the end of the fall term, each student submits a written research proposal following NIH guidelines for either an R01 or K (career development) award. The emphasis in this course is on the quality of the proposed research. The following January or February, students make an oral presentation to the TRANSFORM Advisory Board, summarizing the research proposal. Final grades are awarded after the presentations.

PREREQUISITES

For MS in POR only. Completion of at least 15 credits of the MS-POR curriculum; permission of the instructor.

COURSE LEARNING OBJECTIVES

Students who successfully complete this course will be able to:

- Describe the NIH systems for solicitation, submission, and evaluation of grant proposals.
- Formulate a well-defined clinical or translational research question to be studied in human or animal models.

- Assess the strengths and weaknesses of possible study designs for a given clinical or translational research question.
- Design a research study protocol, incorporating specific aims, the target population, predictor and outcome measures to be assessed, data analysis plan, and timeline for activities.
- Critique a research grant proposal with respect to significance, innovation, and approach.
- Defend your research proposal, recognizing its strengths and limitations.

ASSESSMENT AND GRADING POLICY

This course will be 0 credit for a Pass/Fail grade. Student grades will be based on completion of the following assignments:

Assignment 1..... 10%
Assignment 2.....20%
Assignment 3.....10%
Assignment 4.....40%
Presentations.....20%

It is possible to submit homework assignments late, but there is a penalty of 10% for each day. For example, a student's assignment 2 receives a score of 18 out of 20. If this student submits the assignment within 24 hours after the due date, a 10% of penalty (equivalent to 2 points for assignment 2) will be deducted, so that this student will receive a final score of 16 out of 20 for this assignment. **A cumulative score ≥ 70% will be considered as Pass.**

Note that your final **grade will not be awarded until after the 2025 January/February presentation**. Prior to that, your grade will appear on your transcript as "CP" (credit pending).

Academic integrity is a core value at Mailman. For this reason, the use of generative artificial intelligence (AI) sites, (for example, but not only, Chat GPT) to complete an assignment **is not permitted unless the course instructor has provided clear written instruction about the use of generative AI.**

WRITTEN ASSIGNMENTS

There are 4 written assignments for this course. Please submit all assignments via Canvas. All submissions must be in Word. Here are the assignment details:

<u>ASSIGNMENT #1</u>:

Please submit via Canvas the proposed specific aims page for your proposal, along with the title of your proposal as well as names and email addresses of your mentors. Follow the formatting instructions given below.

- Introductory paragraph
    - What is the topic? What is already known? What are the scientific gaps in knowledge?

    - Why does it matter? (For example, how many people are affected, and how severely? What are the costs?)
    - How will your study address the gaps? How is it innovative?
- Aims
    - What are you proposing to do? State your aims, corresponding hypotheses, primary outcome and exposure variables, population to be studied, and sample size/source of subjects.
- Summary paragraph
    - Describe how completion of these aims will translate to improvements in health.

ASSIGNMENT #2:
Please submit via Canvas the FIRST DRAFT of your proposal, following the formatting instructions given below.

- If you are preparing **an R01 application**, then you must submit the following sections (with page limits):

SECTION	PAGE LIMITS
1. Specific Aims	1 page
2. Research Strategy a. Significance b. Innovation c. Approach (including Preliminary Studies for New Applications, and Progress Report for Renewal/Revision Applications)	12 pages

- If you are preparing **a K23 application**, then you must submit the following sections (with page limits). Note that **BOTH** the candidate information and research strategy sections **must total no more than 12 pages** in your submission:

SECTION	PAGE LIMITS
1. Candidate Information a. Candidate's background b. Career goals and objectives c. Career development/training activities during award period d. Training in the responsible conduct of research	SEE BELOW
2. Specific Aims	1 page
3. Research Strategy a. Significance b. Innovation c. Approach (including Preliminary Studies for New Applications, and Progress Report for Renewal/Revision Applications)	12 pages total, for sections 1 and 3 COMBINED

ASSIGNMENT #3:
Each of you will be assigned, at random, to review the first draft of the proposal submitted by two of your peers. Please prepare a brief (<1 page) written review and submit via Canvas, addressing the following aspects of the study:

- SIGNIFICANCE. Does the project address an important problem or a critical barrier to progress in the field? If the aims of the project are achieved, how will scientific knowledge, technical capability, and/or clinical practice be improved? How will successful completion of the aims change the concepts, methods, technologies, treatments, services, or preventative interventions that drive this field? Has the applicant supplied a sufficient evaluation of the strengths and weaknesses of the data or other justification used to support the application, and described how the proposed research will address any weaknesses or gaps?
- INNOVATION. Does the application challenge and seek to shift current research or clinical practice paradigms by utilizing novel theoretical concepts, approaches or methodologies, instrumentation, or interventions? Are the concepts, approaches or methodologies, instrumentation, or interventions novel to one field of research or novel in a broad sense? Is a refinement, improvement, or new application of theoretical concepts, approaches or methodologies, instrumentation, or interventions proposed?
- APPROACH. Are the overall strategy, methodology, and analyses well-reasoned and appropriate to accomplish the specific aims of the project? Are potential problems, alternative strategies, and benchmarks for success presented? If the project is in the early stages of development, will the strategy establish feasibility and will particularly risky aspects be managed? Has the investigator presented strategies to ensure a robust and unbiased approach, as appropriate for the work proposed?
- OVERALL: What is your overall level of enthusiasm for this application? Note that the overall score need not be an average of the sub-scores, but it should be consistent with them.

ASSIGNMENT #4:
Please submit via Canvas the FINAL DRAFT of your proposal, using the same format as Assignment #2. Please include a cover page, indicating your name, email address, mentors' names and email addresses, title of your proposal, and type of proposal (e.g., R01, K23, etc.). Also include literature cited. These proposals will be reviewed, and comments will be returned to you by mid-January.

MAILMAN SCHOOL POLICIES AND EXPECTATIONS
Students and faculty have a shared commitment to the School's mission, values and oath. mailman.columbia.edu/about/mission-history

Academic Integrity
Students are required to adhere to the Mailman School Conduct and Community Standards, which includes the Code of Academic Integrity. Columbia Mailman and Columbia University take academic integrity very seriously. This instructor and course are no different. Should any

student be suspected of an academic integrity violation, there will be a report submitted to the Center for Student Success & intervention/Student Conduct. After these offices conduct their process, if a student is found responsible for violating an academic integrity policy (see Standards & Discipline/Academic Violations and Student Honor Code & Professional Guidelines), they will be assigned a grade penalty, with a possible outcome being a 0% on the assignment. *[note to faculty: please update this according to your course specific policy]* Please review the university, school, and course policies, as you are responsible for behaving according to the outlined expectations.

Personal Support
Students sometimes experience life challenges that require additional support and connection to resources. If you are experiencing difficult circumstances, please reach out for help and support. Student Support Services in the Office of Student Affairs is poised to connect with students, provide resource referrals, and provide ongoing, non-clinical support. They are a good place to start if you do not know where to turn.

If you would like to make a referral to student support, whether for yourself or someone else, the best way to do so is to fill out the form linked below, and student support will reach out directly. You may also email Meurcie Zignoli at mz3047@cumc.columbia.edu to connect to Student Support Services. Please note that this form is separate from other reporting structures in place for code of conduct violations or Title VI/ Title IX concerns. This form is also not an emergency response mechanism. In case of an emergency, please contact Public Safety at 212-305-7972: Mailman Student Support Team Referral Form (maxient.com)

Disability Access
In order to receive disability-related academic accommodations, students must first be registered with the Office of Disability Services (ODS). Students who have or think they may have a disability are invited to contact ODS for a confidential discussion at 212.854.2388 (V) 212.854.2378 (TTY), or by email at disability@columbia.edu. If you have already registered with ODS, please speak to your instructor to ensure that they have been notified of your recommended accommodations by Meredith Ryer (mr4075@cumc.columbia.edu), Assistant Director of Student Support and Mailman's liaison to the Office of Disability Services.

Bias Incidents
Our community at Columbia University's Mailman School of Public Health is committed to creating an inclusive working, learning, and living environment where all are respected. The occurrence of bias related incidents, involving conduct, speech, or expressions reflecting prejudice are an opportunity for learning and growing as a community.

As part of our efforts to create as inclusive a community as possible, when bias incidents occur at Columbia, we provide an opportunity for those involved to engage in education, advocacy and conversation. In this way, we work to address the incident and minimize the potential for future occurrences. Our community's tools to address bias include a reporting process and the

Bias Incident Resource Team, plus resources within schools and various offices. You can access information about the Bias Reporting Process and FAQs here.

Why Reporting Matters and How to File a Report

Our priority is ensuring that Columbia University is a safe community and workplace where we can learn, live, work and express ourselves. As members of the community, we have a shared responsibility to uphold these standards and report behavior that violates these standards. The reporting options provide Columbia University community members an opportunity to share important information directly with appropriate offices. If you or a member of the community needs support please take the time to complete a report so we may provide support, care, and accountability: https://universitylife.columbia.edu/report